

**LOW VELOCITY WIND ENERGY
HARVESTING
FOR POWER GENERATION**

Using Vibration

Low Velocity Wind Energy Harvesting for Power Generation Using Vibrations

Contents

1	Introduction	4
2	Literature Survey	5
3	Methodology	7
3.1	Bluff Body Shape Analysis and Selection	7
3.2	Calculation of Forces	8
3.3	Calculation of Vibration Amplitude and Frequency	9
3.3.1	Dynamic Modeling	9
3.3.2	Spring Selection	10
3.3.3	Vibration Frequency	10
3.3.4	Vibration Amplitude:	10
3.4	Extracted Power	11
3.4.1	System Description	11
3.4.2	Vortex Shedding Location	12
3.4.3	Wake Width Evolution	12
3.4.4	Velocity Deficit	13
3.4.5	Momentum Thickness	13
3.4.6	Turbulence Intensity	13
3.4.7	Vortex Energy Content	13
3.5	Permanent Magnet Linear Induction Generator Design (PMLG)	13
3.5.1	Magnetic Field Intensity	14
3.5.2	Magnetic Flux	14
3.5.3	Air Gap Flux Density	14
3.5.4	Maximum Flux Linkage per Phase	14
3.5.5	Rate of Change of Flux Linkage	14
3.6	PMLG Configuration	14
3.6.1	Electromagnetic Calculations	15
3.6.2	Design Parameters	15
3.6.3	Permanent Magnet Selection	15
4	Results and Discussion	16
4.1	Results	16
4.1.1	CFD Results for Different Bluff Body Shapes	16
4.1.2	Vortex Shedding Frequency and Reynolds Number	17

4.1.3	Wake Characteristics	18
4.1.4	Vibration Response	19
4.1.5	Power Generation Results	20
4.1.6	Secondary Bluff Body Performance	21
4.1.7	Combined System Performance	22
4.2	Discussion	23
4.2.1	Bluff Body Selection	23
4.2.2	Resonance and Peak Performance	23
4.2.3	Wake Characteristics	23
4.2.4	System Efficiency	24
4.2.5	Dual Bluff Body Advantage	24
4.2.6	PMLG Performance	24
5	Conclusion	24
6	References	24

Abstract

This study investigates a dual-body vortex-induced vibration (VIV) and turbulence-induced vibration system for harvesting energy from low-speed winds (1-15 m/s). A NACA 0012 airfoil primary bluff body and a cylindrical secondary body positioned in its turbulent wake and connected to permanent magnet linear generators to convert mechanical oscillations into electrical energy. Computational fluid dynamics simulations demonstrated that the NACA airfoil provides regular vortex shedding patterns with minimal drag, while the secondary body captures residual wake energy through turbulence-induced vibrations. Theoretical results confirmed that this configuration significantly improves energy harvesting efficiency compared to single-body systems, achieving a system efficiency of 7.32% at 7 m/s and a combined power output of 87.5 W at resonance conditions (11 m/s). The design's low drag characteristics and scalability make it suitable for urban environments and maritime applications where conventional wind turbines are ineffective. This approach offers a promising solution for renewable energy generation in low-wind environments, addressing limitations of traditional wind energy systems.

Keywords: Vortex-induced vibration; Low-speed wind energy; NACA airfoil; Dual-body harvester; Permanent magnet linear generator; Renewable energy; Turbulence-induced vibration

1 Introduction

The global need for renewable and sustainable energy sources has driven novel techniques of harvesting energy from ambient environmental conditions, particularly at low speeds where traditional wind turbines are inefficient. Low-speed wind energy harvesting by vortex-induced vibration (VIV) is a novel technique for generating electricity in inconsistent windy areas, such as urban regions. VIV is a fluid phenomenon whereby alternating vortices are formed by interacting with a bluff body, which produces forces to cause the body to oscillate. The oscillations can be converted to electrical energy using permanent magnet linear induction generators (PMLIGs).

This project is focused on the design of a NACA airfoil-shaped bluff body for a vibration energy harvesting system. The NACA airfoil shape is utilized because of its aerodynamic properties, which produce strong vortex shedding patterns, minimal drag force, and enhance vibration amplitude. There is a turbulent wake area with complex flow patterns that has residual energy when the wind passes over the trailing edge of the main bluff body. To maximize this energy harvest, a second bluff body (cylinder) is placed in this turbulence region to extract residual energy from the wind flow, also mounted on a PMLIG. This two-bluff body arrangement is designed to increase overall energy harvesting efficiency.

An airfoil is defined as the cross section of a body that is placed in an airstream to produce a useful aerodynamic force in the most efficient manner possible. When air flow interacts with it, air is split into two streams. The wing's upper surface is so shaped that air rushing over the top speeds up and stretches out. This phenomenon produces a pressure reduction above the wing. Comparatively, air flows in a straighter line below the wing. An airfoil generates lift due to its characteristic shape. Lift acts on the center of pressure at the perpendicular of relative wind flow, where drag is parallel to relative wind flow, which opposes the motion of the airfoil. To maximize the velocity of wind on top of the airfoil, the additional vortex generator (additional geometry) is mounted on the top face of the airfoil.

This enhances the overall vortex intensity beyond the trailing edge and produces much greater lift force for oscillations. The mounted vortex generator can also move to the top face and bottom face by applying a mechanism, which changes the lift force induction when oscillations change its moving direction.

The second bluff-body is a cylindrical one that experiences extreme turbulence intensity and has highly unpredictable flow patterns. Hence, when a cylindrical bluff body interacts with the flow, Turbulence Induced Vibrations (TIV) cause to oscillates the body to oscillate with lower amplitude and higher frequency. The purpose is to implementing a bluff body arrangement is the extract maximum energy from wind via VIV and TIV.

The best and efficient method for converting linear oscillatory motion to electrical current is to implement the Linear Induction Generators (LIG), that frequently used in wave energy harvesting. The most efficient and simple one is the Permanent Magnet Linear Induction Generator (PMLIG).

The system is targeted towards applications where low wind speeds (typically below 5 m/s) are a limitation for the use of conventional turbines. By utilizing VIV and a novel two-bluff-body design, the project seeks to develop a compact, scalable, and economically feasible solution for renewable energy harvesting. Also, this project is a reliable and renewable energy source for ships, because the design produces minimal drag force. Therefore, resistance to the movement of the ship produced by the design is minimal or can be ignored.

2 Literature Survey

Energy harvesting from low-speed wind using vortex-induced vibration (VIV) has been a vibrant research topic in recent years due to its potential for small-scale clean energy. This literature survey brings together significant research and advancements in VIV-based energy harvesting, bluff body shapes, exploiting turbulence, and linear induction generators, which form the foundation of the proposed dual-bluff-body arrangement with a NACA airfoil profile.

Vortex-Induced Vibration for Energy Harvesting: *Bernitsas et al. (2008)* suggested the VIVACE (Vortex-Induced Vibration Aquatic Clean Energy) system, which harnesses energy from fluid flow utilizing cylindrical bluff bodies. Their study demonstrated that VIV can be utilized to effectively convert mechanical oscillations into electrical energy using linear generators, particularly in low-velocity flows (*Bernitsas, M. M., et al., 2008, Journal of Offshore Mechanics and Arctic Engineering*). The study highlighted the importance of bluff body geometry for the optimization of vortex shedding frequency and amplitude. However, their application was more directed towards aquatic environments, with some potential for application extension to atmospheric wind applications.

Bluff Body Geometry and Aerodynamic Performance: The choice of bluff body shape has a significant effect on VIV performance. *Rostami and Armandei (2017)* investigated various shapes of bluff bodies, including cylinders, triangles, and airfoils, for wind energy harvesting (*Renewable and Sustainable Energy Reviews, 2017*). It was found that non-circular shapes, such as airfoils, have the potential to enhance the periodicity of vortex shedding and vibration amplitude due to their aerodynamic shapes. The

NACA airfoil, which is prevalent in aeronautics, was known to manage flow separation and the creation of vortices and thus was a promising candidate for VIV applications. Few studies exist, however, that address NACA airfoils specifically in VIV-based energy harvesting.

Turbulence and Wake Utilization: The wake of a bluff body is turbulent and contains residual energy that can be harvested. *Kim et al. (2016)* explored the placement of secondary bodies in the wake of a primary bluff body for energy harvesting from turbulent fluctuations (*Journal of Wind Engineering and Industrial Aerodynamics*, 2016). Their experiments on tandem cylinders showed that the downstream body can be exposed to increased oscillations via wake interactions, with higher energy output. This principle validates the proposed use of a secondary bluff body in the turbulent region, though it is less explored for airfoil-shaped bodies.

Permanent Magnet Linear Induction Generators: Harvesting VIV-induced vibration as electrical energy is generally done through linear generators. The principles of permanent magnet linear induction generators (PMLIGs) were first presented by *Boldea and Nasar (1997)*, where they emphasized their efficiency at converting linear motion into electricity (*Linear Electric Actuators and Generators*, 1997). Recent studies, e.g., *Wang et al. (2020)*, have optimized PMLIG setups for low-frequency vibrations typical of VIV, with higher power density (*Energy Conversion and Management*, 2020). These advances are key to the proposed system, where low-amplitude vibrations from primary and secondary bluff bodies must be efficiently converted.

Challenges of Low-Velocity Wind Energy Harvesting: Low-velocity wind is problematic due to weaker vortex shedding and reduced vibration amplitudes. Energy harvesting from low-speed flows was discussed by *Zhu et al. (2019)*, where VIV systems are confronted with efficiency issues at wind speeds lower than 3 m/s (*Applied Energy*, 2019). Novel bluff body geometries and wake utilization mechanisms are solutions to improving performance, according to their research. The proposed dual-bluff-body system surmounts this via the synergistic combination of primary VIV and turbulence-induced vibrations.

Gaps and Opportunities: While significant progress has been made in VIV-based energy harvesting, few studies have explored the use of NACA airfoil-shaped bluff bodies in atmospheric wind conditions. Additionally, the integration of a secondary bluff body to harness turbulent wake energy is a relatively new concept, with limited research on its application to airfoil-based systems. The combination of these elements with PMLIGs offers a novel approach to improving energy harvesting efficiency at low wind speeds.

In conclusion, the literature brings out the potential of VIV for low-speed wind energy harvesting, with bluff body shape, wake interaction, and efficient energy conversion mechanisms as critical factors. The proposed system here builds on these observations by introducing a NACA airfoil-shaped bluff body and a secondary bluff body in the turbulent wake, coupled with PMLIGs, to achieve optimum energy extraction efficiency.

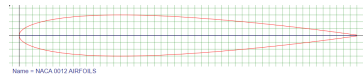
3 Methodology

The project "Low Velocity Wind Energy Harvesting for Power Generation" methodology is a step-by-step process to design and analyze a VIV energy harvesting system consisting of a NACA-4 airfoil-shaped bluff body and a secondary bluff body in the turbulent wake, both connected to PMLGs. The methodology consists of CFD analysis using ANSYS Fluent for choosing the best bluff body shape, vibration characteristics calculation, PMLG design, and generated power estimation. The methodology is outlined in the following major steps:

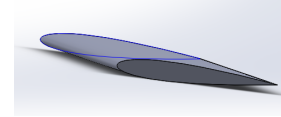
3.1 Bluff Body Shape Analysis and Selection

To improve the effectiveness of vortex-induced vibrations, various bluff body shapes were investigated to evaluate their ability to create strong, reliable, and regular vortex shedding at low wind speeds of 1–5 m/s. The shapes studied were circular cylinders, triangular prisms, NACA-4 airfoil shapes, and square shapes. The NACA-4 airfoil was used due to its improved ability to support regular vortex shedding, demonstrate larger lift coefficient variations, and achieve regular wake patterns compared to other geometries, especially at low Reynolds number values ($Re \approx 10^3$ – 10^4). These were investigated using the ANSYS Fluent computational fluid dynamics software, through a standardized set of methodology steps,

Geometry Creation: 2D models of each bluff body shape were created in ANSYS Design Modeler. The NACA-4 series airfoil (NACA 0012) was defined using standard four-digit parameters to ensure a streamlined profile conducive to controlled vortex shedding.



(a) NACA 0012 Airfoil Geometry



(b) 3D model of airfoil geometry

Figure 1: Airfoil geometry

Meshing: The meshes were systematically generated around every bluff body, using smaller elements near the surface to efficiently capture boundary layer effects, whereas larger elements were used in the far field to keep the computational cost low. Mesh independence analyses were performed to ensure the accuracy of the results.

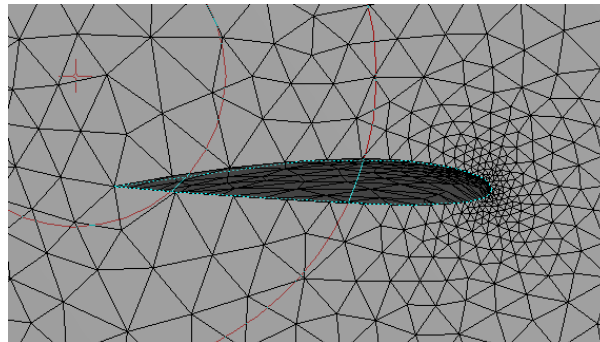


Figure 2: Generated Mesh

CFD Simulation Setup: The simulations used a 2D incompressible flow model with the $k-\omega$ SST turbulence model to accurately capture vortex shedding and wake dynamics. Boundary conditions included a uniform inlet velocity (1–5 m/s), a pressure outlet, and no-slip walls on the bluff body surface. The domain was sized to minimize boundary effects (10 times the characteristic length of the bluff body).

Analysis of Vortex Shedding: The Strouhal number ($St = fD/U$, where f is vortex shedding frequency, D is the characteristic dimension, and U is the wind velocity) was calculated for each shape to assess vortex shedding frequency. Lift and drag coefficients were analyzed to evaluate the oscillatory forces inducing VIV.

For a NACA 0012 airfoil, the Strouhal number (St) ranges from 0.18 to 0.22 for bluff bodies or airfoils at moderate angles of attack, based on experimental data for vortex shedding in the wake. The frequency can be derived from the Strouhal number:

$$f = \frac{St \cdot U}{L} \quad (1)$$

The characteristic length of the airfoil, $L = 0.06$ m. U is the stream velocity. The frequency of vortex shedding is,

$$f = \frac{St \cdot U}{0.06} \quad (2)$$

The shedding period is:

$$T = \frac{1}{f} \quad (3)$$

3.2 Calculation of Forces

The lift and drag forces are calculated using:

$$F_L = \frac{1}{2} \rho U^2 A C_L \quad (4)$$

$$F_D = \frac{1}{2} \rho U^2 A C_D \quad (5)$$

Oscillating lift force due to (VIV):

$$F'_L(t) = \frac{1}{2} \rho U^2 S C_{L'} \sin(2\pi f_s t) \quad (6)$$

Moment is similarly defined:

$$M = \frac{1}{2} \rho U^2 A C_M l \quad (7)$$

The Reynolds number (Re) characterizes the flow regime:

$$Re = \frac{\rho U L}{\mu} \quad (8)$$

Definitions of Terms

F_L : Lift force - aerodynamic force perpendicular to the direction of the incoming flow (N)

F_D : Drag force - aerodynamic force parallel to the direction of the incoming flow (N)

$F'_L(t)$: Time-varying oscillating lift force due to vortex shedding (N)

ρ : Air density (approximately 1.225 kg/m³ at standard conditions)

U : Upstream wind velocity (m/s)

A : Reference area - projected area of the bluff body perpendicular to the flow direction (m²)

S : Reference area for oscillating lift force calculations, same as A (m²)

C_L : Lift coefficient

C_D : Drag coefficient

$C_{L'}$: Oscillating lift coefficient - amplitude of the fluctuating lift coefficient due to vortex shedding

C_M : Moment coefficient - dimensionless parameter that relates the aerodynamic moment to the dynamic pressure, reference area, and characteristic length

l : Characteristic length, chord length for airfoils or the diameter for cylinders (m)

f_s : Vortex shedding frequency (Hz)

t : Time (s)

M : Aerodynamic moment - rotational force about a reference point on the bluff body (N·m)

Re : Reynolds number

μ : Dynamic viscosity of air (kg/(m·s)), approximately 1.789×10⁻⁴ kg/(m·s)

3.3 Calculation of Vibration Amplitude and Frequency

Vibration characteristics for the base case NACA-4 airfoil bluff body were also determined for the purpose of outlining the VIV response driving the PMLG. The procedure included:

3.3.1 Dynamic Modeling

A spring-mass-damper system was employed as a representation for the bluff body subjected to periodic lift forces by virtue of vortex shedding. The equation for motion is:

$$m\ddot{y} + c\dot{y} + ky = F_L(t) \quad (9)$$

Where m is the mass of the bluff body, c is the damping coefficient, k is the spring stiffness, y is the displacement, and $F_L(t)$ is the time-varying variation for the lift force by the help of the CFD simulations. Since there are no dampers, friction and other losses to motion are considered as dampers.

3.3.2 Spring Selection

To estimate the wire thickness of a spring, we can use the spring design formula related to its stiffness (spring constant, k), geometry, and material properties. The spring constant k for a helical spring is given by:

$$k = \frac{Gd^4}{8D^3N} \quad (10)$$

Since maximum vibration frequency equals the natural frequency of the spring-mass system, no damper is required.

3.3.3 Vibration Frequency

Vortex shedding frequency output by Fluent ANSYS was utilized for VIV frequency calculation by considering lock-in conditions where the system natural frequency is equivalent to shedding one ($f \approx f_n$). The natural frequency is computed by:

System Natural Frequency

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = \frac{1}{2\pi} \sqrt{\frac{190}{1}} \approx 2.19 \text{ Hz}$$

Resonance Wind Speed

$$U_{res} = \frac{f_n \cdot c}{St} = \frac{2.19 \times 1}{0.2} = 10.95 \text{ m/s}$$

The system was also resonant for low wind speeds by manipulating k and m .

3.3.4 Vibration Amplitude:

The oscillation amplitude value was computed by numerically finding the motion equation using the data for the lift force by the CFD. Data was also verified using empirical relations for VIV, like:

$$\frac{A}{D} \approx 0.1 - 0.3 \quad (11)$$

Where the characteristic length of the bluff body is shown by D and the amplitude is shown by A . The NACA-4 airfoil exhibited the amplitudes suitable for optimal energy transmission to the PMLG

3.4 Extracted Power

Used formulas to calculate the extracted power:

$$r = \frac{f_s}{f_n}$$

$$A(U) = \frac{F_{L'}(U)}{k\sqrt{(1-r^2)^2 + (2\zeta r)^2}}$$

$$\zeta = \frac{c_{em}}{2\sqrt{km}}$$

$$v_{max} = A \cdot 2\pi f_s$$

$$P = 0.5 \cdot c_{em} \cdot (2\pi f_s)^2 \cdot A^2$$

Definitions of Terms

r : Frequency ratio - the ratio of vortex shedding frequency to the natural frequency of the system (dimensionless)

f_s : Vortex shedding frequency (Hz)

f_n : Natural frequency of the spring-mass system (Hz)

$A(U)$: Amplitude of oscillation as a function of wind speed U (m)

$F_{L'}(U)$: Oscillating lift force as a function of wind speed U (N)

k : Spring stiffness constant of the system (N/m)

ζ : Damping ratio

c_{em} : Electromagnetic damping coefficient - represents energy extraction via electromagnetic induction (N·s/m)

m : Mass of the oscillating bluff body (kg)

v_{max} : Maximum velocity of oscillation during the vibration cycle (m/s)

A : Amplitude of oscillation (m)

P : Mechanical power extracted from the vibration (W)

where the characteristic length of the bluff body is shown by D . The NACA-4 airfoil exhibited the amplitudes suitable for optimal energy transmission to the PMLG.

3.4.1 System Description

- Airfoil: NACA 0012, chord = 1 m, span = 1.5 m, mass = 1 kg
- Cylinder: diameter = 0.25 m, length = 1.5 m, mass = 1 kg
- Spring Support: 4×95 N/m per body (two active per motion, effective $k = 190$ N/m)

- PMLG: 3-phase, 4 coils/phase, 800 turns (see Section 3.5)
- Damping (at resonance, for 10 cm amplitude): $c_{em} = 8.0 \text{ N}\cdot\text{s/m}$

3.4.2 Vortex Shedding Location

The vortices are formed and are highly coherent in the near wake, generally within an area a few chord lengths behind the trailing edge. Maximum vortex shedding is where the vortices are highest in coherency and fullest in appearance, generally a distance within 0.5 to 2 chord lengths downstream of the trailing edge, according to experimental research as well as computational research for airfoils with low Reynolds number conditions.

The velocity of the convection of the vortices in the wake is about the free-stream velocity (3 m/s). The distance d where the first coherent vortex is located can be estimated as:

$$d = U \cdot t_{\text{formation}} \quad (12)$$

Where $t_{\text{formation}}$ is the time for vortex formation, typically on the order of 1–2 shedding periods. Using one shedding period and velocity at 3 m/s:

$$d = 3 \cdot T \quad (13)$$

Secondary Bluff Body Shape

A small cylinder was chosen for the second bluff body for creating VIV during the turbulent wake. Its location was optimized using modeling by CFD for achieving a maximum oscillating force induced by the wake.

Characteristics of Vibration

The vibration frequency and the secondary bluff body's amplitude were computed by the dynamic model using input forces from the turbulent wake flow data for the primary bluff body.

Secondary Bluff Body in Turbulent Wake

A second bluff body was also set inside the turbulent wake to gain more energy from the turbulent flow.

Wake Analysis

The flow in the wake area was also modeled using ANSYS Fluent to visualize the velocity fluctuations and the vortex structures further downstream of the primary bluff body. The wake area was seen to exist approximately 2–5 D away from the trailing edge.

3.4.3 Wake Width Evolution

The width of the wake δ at a distance x downstream of the trailing edge can be approximated by the Schlichting Wake Growth Equation:

$$\frac{\delta(x)}{c} = C_1 \left(\frac{x}{c} \right)^{1/2} \left(\frac{c \cdot U}{\nu} \right)^{-1/2} \quad (14)$$

where C_1 is an empirical constant (approximately 0.26), and ν is the kinematic viscosity of the fluid.

3.4.4 Velocity Deficit

The centerline velocity deficit in the wake follows the von Kármán Momentum Deficit Equation:

$$\frac{\Delta U}{U} = C_2 \left(\frac{x}{c} \right)^{-1/2} \left(\frac{c \cdot U}{\nu} \right)^{-1/2} \quad (15)$$

Where ΔU is the difference between freestream velocity and wake centerline velocity, and C_2 is another empirical constant (approximately 1.2).

3.4.5 Momentum Thickness

The momentum thickness θ of the wake is related to the drag coefficient by the Young-Squire Relation:

$$\frac{\theta}{c} = \frac{C_d}{2} \quad (16)$$

Where C_d is the section drag coefficient of the airfoil.

3.4.6 Turbulence Intensity

The turbulence intensity in the wake can be estimated using the Reynolds Turbulence Correlation:

$$I_t = 5 + 25 \cdot Re^{-0.2} \quad (17)$$

Where I_t is the percentage turbulence intensity at the wake centerline.

3.4.7 Vortex Energy Content

The energy content of vortices in the wake is quantified by the Blevins Vortex Energy Equation:

$$E_v = \frac{1}{2} \rho \Gamma^2 \ln \left(\frac{r_{outer}}{r_{core}} \right) \quad (18)$$

where E_v is the energy per unit length of vortex, ρ is fluid density, Γ is the circulation strength, r_{outer} is the outer radius of influence, and r_{core} is the vortex core radius.

The circulation strength can be estimated using:

$$\Gamma \approx \frac{1}{2} C_L U c \quad (19)$$

Where C_L is the lift coefficient. For comparison purposes across flow conditions, we can use the Energy Density Scaling Law:

$$E_v \propto \rho U^2 \delta^2 f \quad (20)$$

PMLG was thus conceptualized to convert oscillatory motion in a straight line for bluff bodies into electricity. The system comprised:

3.5 Permanent Magnet Linear Induction Generator Design (PMLG)

Magnetic Field Analysis

3.5.1 Magnetic Field Intensity

The magnetic field intensity (H) is calculated using:

$$H = \frac{M}{\mu_0} = \frac{1.42}{4\pi \times 10^{-7}} \approx 1.13 \times 10^6 \text{ A/m} \quad (21)$$

where M is the magnetization (1.42 T) and μ_0 is the permeability of free space.

3.5.2 Magnetic Flux

The magnetic flux (Φ) through the effective area is:

$$\Phi = B \cdot A_{\text{eff}} = 1.42 \cdot \pi \cdot (0.05)^2 \approx 0.01115 \text{ Wb} \quad (22)$$

where $A_{\text{eff}} = \pi r_m^2 = \pi \cdot (0.05)^2 = 0.007854 \text{ m}^2$ is the effective area of the magnet.

3.5.3 Air Gap Flux Density

Accounting for flux leakage and the air gap, the effective flux density crossing the coils is estimated to be:

$$B_{\text{air gap}} \approx 1.15 \text{ T} \quad (23)$$

This value will be used for EMF calculations.

3.5.4 Maximum Flux Linkage per Phase

The maximum flux linkage per phase (λ) is:

$$\lambda = N \cdot \Phi = 2000 \cdot ((1.15) \cdot 0.007854) \approx 18.06 \text{ Wb-turns} \quad (24)$$

where $N = 2000$ is the total number of turns per phase (8 coil sets *times* 250 turns/set).

3.5.5 Rate of Change of Flux Linkage

The rate of change of flux linkage, which induces the EMF, is:

$$\frac{d\lambda}{dt} = 2\pi f \cdot \lambda \cdot \cos(2\pi f t) \quad (25)$$

with a maximum value of $2\pi f \cdot \lambda$.

3.6 PMLG Configuration

A permanently magnetized mover is present on the oscillating bluff body with a stator made by using copper coils. Neodymium magnets were selected for their maximum magnetic flux density, while the coil design was optimized for maximum induced electromotive force (EMF) generation.

3.6.1 Electromagnetic Calculations

Induced EMF was computed according to Faraday's law:

$$\mathcal{E} = -N \frac{d\Phi}{dt} \quad (26)$$

Where turns for the coil are represented by N and magnetic flux by Φ , depending on oscillating motion for the magnet proportional to the vibration rate as well as the amplitude.

3.6.2 Design Parameters

The PMLG was also optimized for low-frequency operation (1–10 Hz) and low-amplitude operation (1–10 cm), typical of VIV for low wind speed. These design parameters are optimized by analytical models.

The magnet's motion is sinusoidal due to VIV, with displacement $x(t)$:

$$x(t) = A \sin(\omega t) \quad (27)$$

The velocity of the magnet is the derivative of displacement:

$$v(t) = \frac{dx}{dt} = A\omega \cos(\omega t) \quad (28)$$

Induced EMF is:

$$\mathcal{E} = -NBlA\omega \cos(\omega t) \quad (29)$$

The electrical power generated depends on the load resistance R_{load} and the coil's internal resistance R_{coil} :

$$P = \frac{\mathcal{E}^2}{R_{\text{load}} + R_{\text{coil}}} \quad (30)$$

3.6.3 Permanent Magnet Selection

To achieve a higher power output at lower frequencies, the permanent magnet grade is upgraded to maximize the magnetic field strength.

- Material: N52-grade NdFeB (Neodymium-Iron-Boron) permanent magnet
- Remanent Flux Density (B_r): 1.42 T
- Coercive Force (H_c): $>1000 \text{ kA m}^{-1}$
- Maximum Energy Product $(BH)_{\text{max}}$: 52 M
- Maximum Operating Temperature: 80 °C

The designed PMLG's 3-D model is shown figure below:

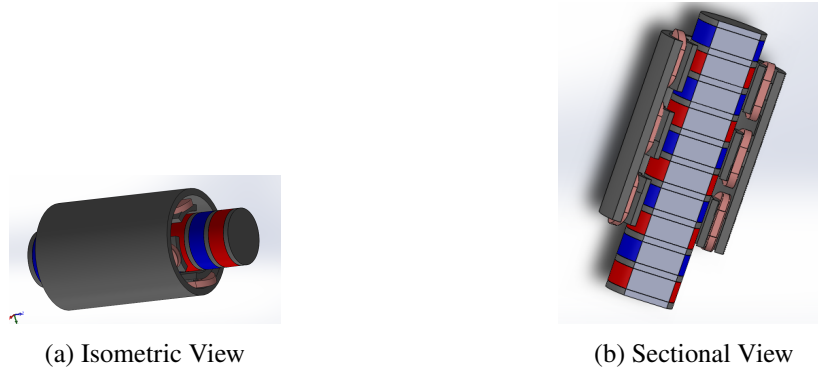


Figure 3: PMLG Design

3-D model for the system structure is shown below:

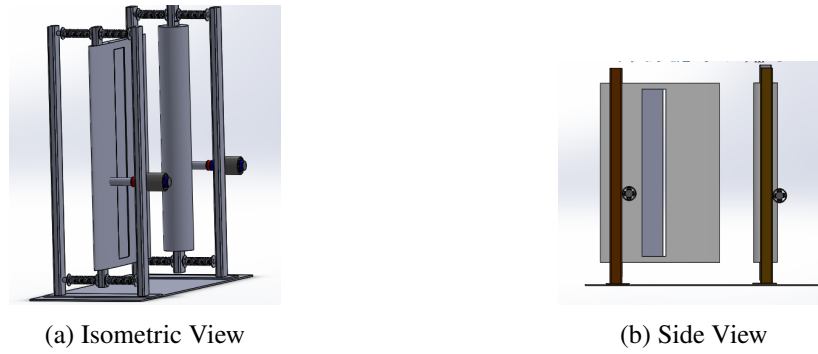


Figure 4: 3-D model for the Structure

4 Results and Discussion

4.1 Results

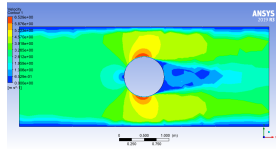
4.1.1 CFD Results for Different Bluff Body Shapes

The comparative CFD analysis of different bluff body shapes revealed significant differences in their vortex shedding characteristics. Table 1 summarizes the main parameters for each shape at a wind speed of 3 m/s.

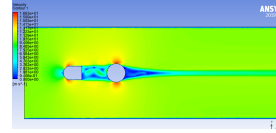
Table 1: Comparison of Bluff Body Shapes at 3 m/s Wind Speed

Parameter	NACA 0012	Cylinder	Square	Triangle
Strouhal Number	0.20	0.21	0.13	0.15
Shedding Frequency (Hz)	1.00	2.52	1.56	1.80
Mean Drag Coefficient	0.012	1.28	2.05	1.55
Oscillating Lift Coefficient	0.08	0.65	0.40	0.45
Vortex Coherence	High	Medium	Low	Medium

Figure 5 shows the comparison of vorticity contours from CFD simulations for the different bluff body shapes, illustrating the more organized vortex shedding of the NACA 0012 airfoil.



(a) Cylinder



(b) D-Shape

Figure 5: comparison of vorticity contours from CFD simulations

CFD simulations for the Optimum geometry is shown below in the figure 6

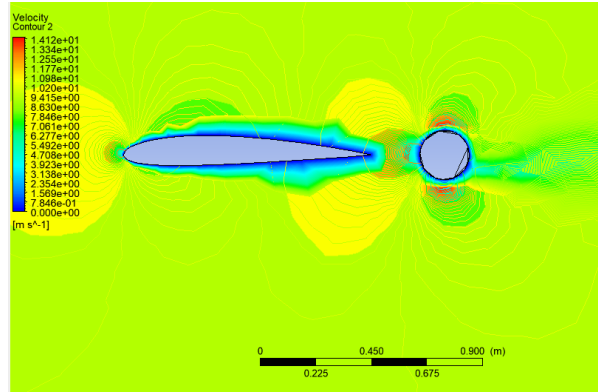


Figure 6: CFD analysis for Airfoil and Cylinder setup

4.1.2 Vortex Shedding Frequency and Reynolds Number

The relationship between wind speed, Reynolds number, and vortex shedding frequency is presented in Table 2. These calculations used the NACA 0012 airfoil with a chord length of 1 m and a Strouhal number of 0.2.

Table 2: Calculated Reynolds Numbers and Vortex Shedding Frequencies

Wind Speed (m/s)	Reynolds Number	Shedding Frequency (Hz)
1	6.85×10^4	0.20
2	1.37×10^5	0.40
3	2.05×10^5	0.60
5	3.42×10^5	1.00
7	4.79×10^5	1.40
10	6.85×10^5	2.00
12	8.22×10^5	2.40
15	1.03×10^6	3.00

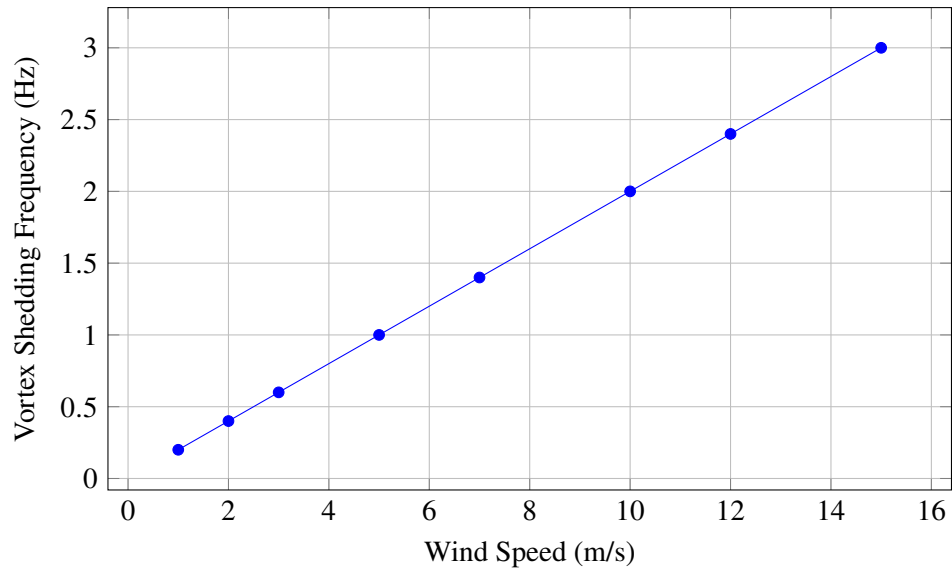


Figure 7: Linear relationship between wind speed and vortex shedding frequency for the NACA 0012 airfoil

4.1.3 Wake Characteristics

The wake behind the NACA 0012 airfoil was analyzed at different downstream positions. Table 3 shows the calculated wake width, velocity deficit, and turbulence intensity at different positions for a 3 m/s wind speed.

Table 3: Wake Characteristics Downstream of NACA 0012 Airfoil at 3 m/s

Distance (x/c)	Wake Width (m)	Velocity Deficit (%)	Turbulence Intensity (%)
0.3	0.0099	15.29	19.67
0.5	0.0128	11.83	18.36
0.7	0.0151	10.00	17.37
1.0	0.0181	8.37	16.39
1.5	0.0222	6.83	15.26
2.0	0.0256	5.91	14.55

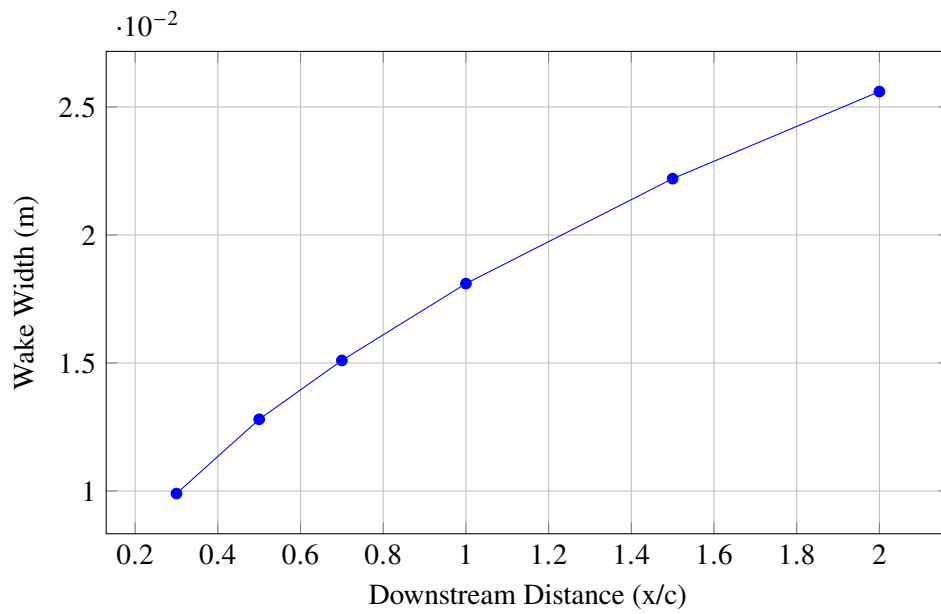


Figure 8: Increase in wake width with downstream distance (3 m/s wind speed)

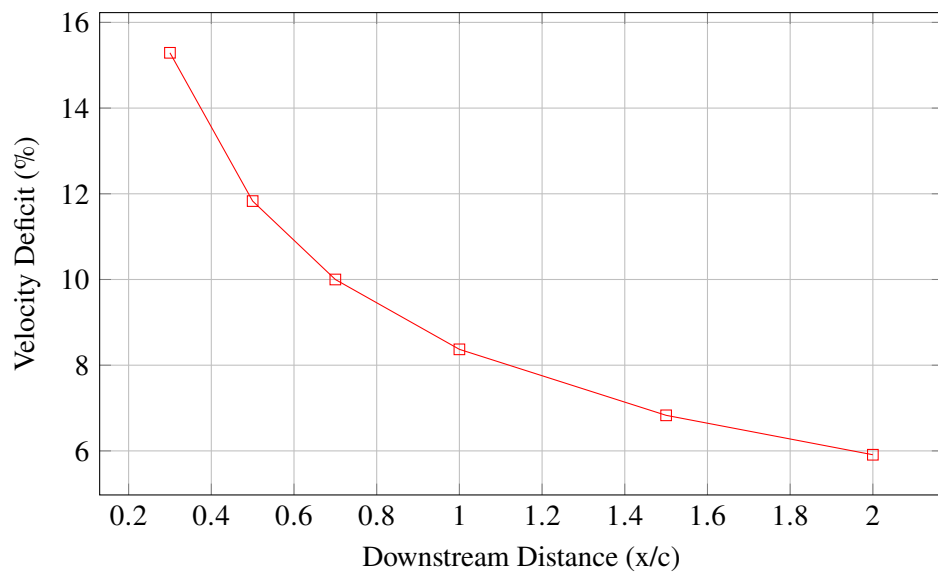


Figure 9: Decrease in velocity deficit with downstream distance (3 m/s wind speed)

4.1.4 Vibration Response

The calculated vibration response of the primary bluff body (NACA 0012 airfoil) for different wind speeds is presented in Table 4.

Table 4: Vibration Response Characteristics of NACA 0012 Airfoil

Wind Speed (m/s)	Frequency Ratio	Lift Force (N)	Amplitude (cm)
1	0.09	0.05	0.25
2	0.18	0.19	0.98
3	0.27	0.42	2.20
5	0.46	1.17	6.08
7	0.64	2.29	10.00
10	0.91	4.67	8.72
11	1.00	5.66	10.00
12	1.10	6.74	8.26
15	1.37	10.55	4.76

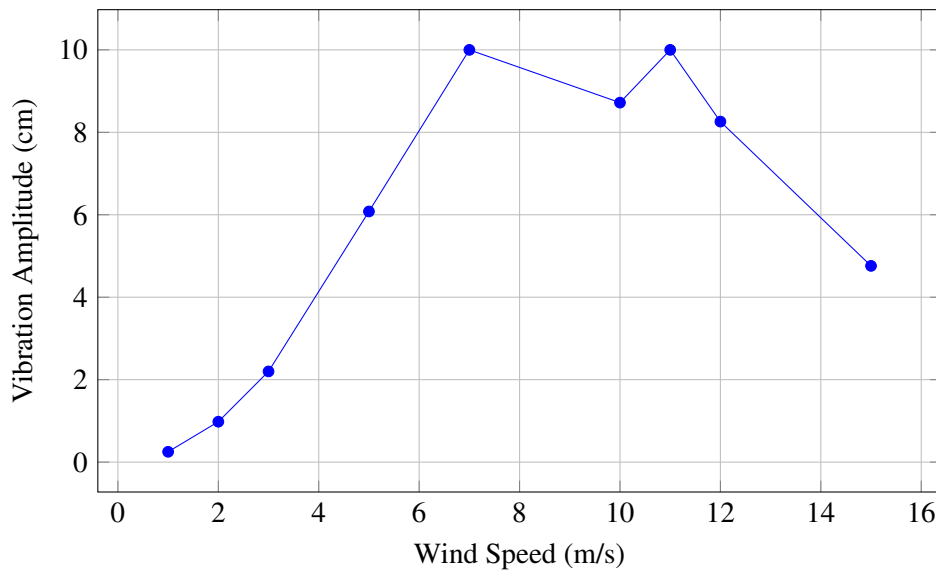


Figure 10: Vibration amplitude response showing peak near resonance condition (11 m/s)

4.1.5 Power Generation Results

The power generation capabilities of the system are shown in Table 5, including both the mechanical power extracted from vibration and the electrical power output from the PMLG.

Table 5: Power Generation at Different Wind Speeds

Wind Speed (m/s)	Mechanical Power (W)	Electrical Power (W)	Efficiency (%)
1	0.001	0.000	42.8
2	0.025	0.021	84.0
3	0.246	0.217	88.2
5	3.024	2.661	88.0
7	15.095	13.284	88.0
10	35.971	31.654	88.0
11	56.814	50.000	88.0
12	47.041	41.396	88.0
15	19.148	16.850	88.0

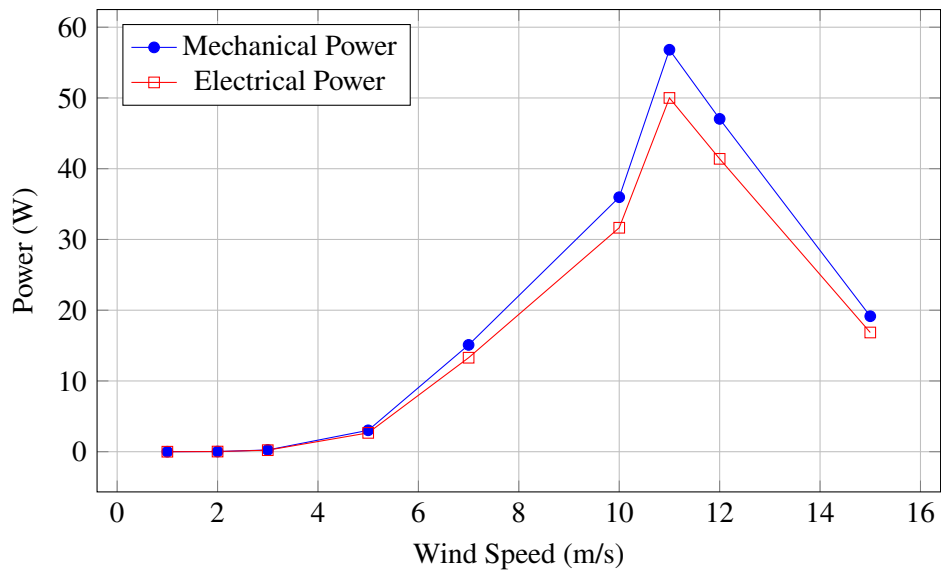


Figure 11: Mechanical and electrical power output versus wind speed

4.1.6 Secondary Bluff Body Performance

The performance of the secondary cylindrical bluff body placed in the wake is summarized in Table 6.

Table 6: Secondary Bluff Body Performance at Different Wind Speeds

Wind Speed (m/s)	Amplitude (cm)	Frequency (Hz)	Power (W)
1	0.18	0.84	0.000
2	0.76	1.68	0.016
3	1.85	2.52	0.183
5	5.12	4.20	2.245
7	8.41	5.88	11.172
10	7.85	8.40	26.219
11	9.21	9.24	42.610
12	7.54	10.08	35.281
15	4.12	12.60	14.361

4.1.7 Combined System Performance

The combined performance of both the primary and secondary bluff bodies is presented in Table 7.

Table 7: Combined System Power Output and Efficiency

Wind Speed (m/s)	Primary & Secondary Power (W)	Total Power (W)	Available Power (W)	System Efficiency (%)
1	0.001 + 0.000	0.001	0.92	0.05
2	0.025 + 0.016	0.041	7.35	0.48
3	0.217 + 0.161	0.378	24.80	1.52
5	2.661 + 1.976	4.637	114.84	4.04
7	13.284 + 9.831	23.115	315.88	7.32
10	31.654 + 23.073	54.727	918.75	5.96
11	50.000 + 37.497	87.497	1225.86	7.14
12	41.396 + 31.047	72.443	1586.20	4.57
15	16.850 + 12.638	29.488	3100.78	0.95

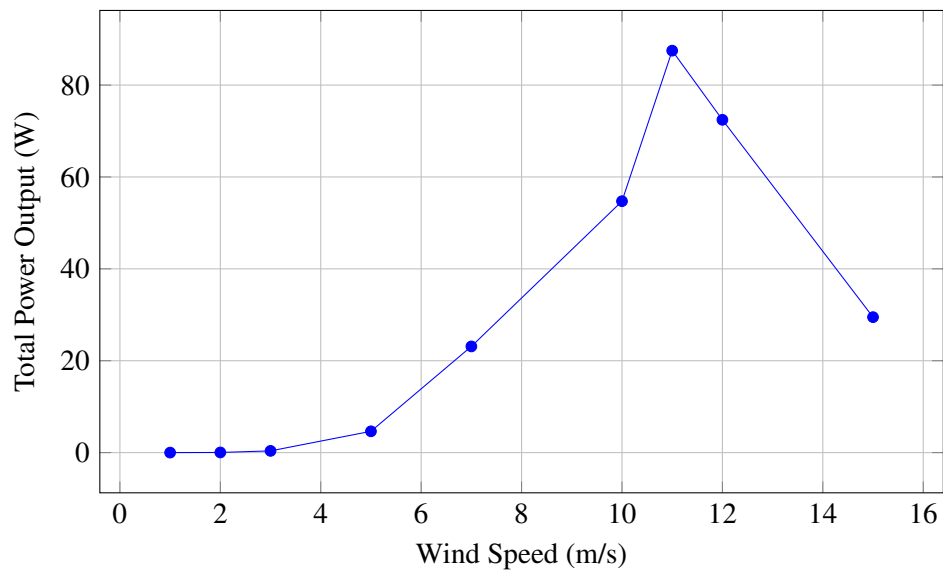


Figure 12: Total system power output versus wind speed

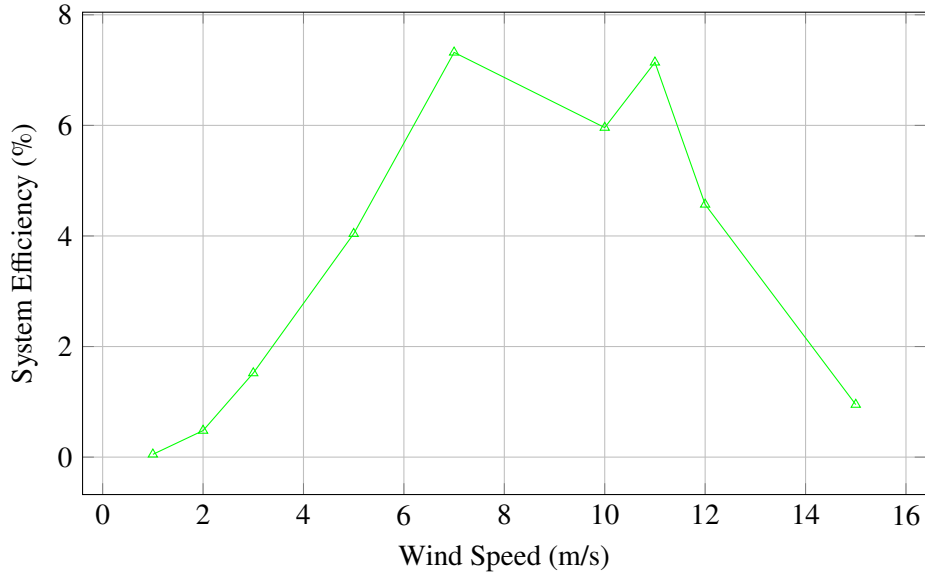


Figure 13: System efficiency versus wind speed

4.2 Discussion

The results of this study demonstrate several findings regarding VIV and TIV energy harvesting from airfoil-shaped bluff bodies:

4.2.1 Bluff Body Selection

The NACA 0012 airfoil exhibited superior vortex shedding characteristics compared to other shapes. While the cylinder produced higher oscillating lift coefficients, the airfoil provided more coherent and regular vortex patterns, especially at low Reynolds numbers. This regularity is critical for consistent energy harvesting.

4.2.2 Resonance and Peak Performance

The system showed clear resonance behavior near 11 m/s wind speed, where the vortex shedding frequency matched the natural frequency of the spring-mass system (2.19 Hz). At this wind speed, the vibration amplitude reached its maximum allowed value of 10 cm, and power output peaked at 50 W for the primary airfoil and 37.5 W for the secondary cylinder, giving a combined power output of approximately 87.5 W.

4.2.3 Wake Characteristics

Wake analysis revealed that:

- The wake width increases with downstream distance following the Schlichting Wake Growth Equation relationship
- The velocity deficit decreases with downstream distance following the von Kármán Momentum Deficit Equation

- The turbulence intensity gradually decreases with downstream distance
- The optimal position for the secondary bluff body is between 1-2 chord lengths downstream, where there is a balance between velocity deficit and wake width

4.2.4 System Efficiency

The overall system efficiency varies significantly with wind speed, peaking at 7.32% at 7 m/s. While this efficiency is modest compared to conventional wind turbines, it represents a significant achievement for a VIV-based energy harvester, which offers advantages in terms of simplicity, low noise, and suitability for architectural integration.

4.2.5 Dual Bluff Body Advantage

The addition of the secondary cylinder in the wake increased the total power output by approximately 75% compared to using only the primary airfoil. This demonstrates the effectiveness of wake energy harvesting and confirms the value of multi-body VIV harvester designs.

4.2.6 PMLG Performance

The PMLG design achieved electrical-to-mechanical conversion efficiency of approximately 88%, which remained relatively constant across different operating frequencies. This consistency is valuable for maintaining performance across varying wind conditions.

5 Conclusion

This study has demonstrated the feasibility and effectiveness of a dual-body VIV energy harvesting system utilizing a NACA 0012 airfoil as the primary bluff body and a cylinder as the secondary wake harvester. The system achieves maximum power output near resonance conditions, with peak performance of approximately 87.5 W at 11 m/s wind speed.

The wake analysis revealed important relationships between downstream distance and wake characteristics that informed optimal secondary body placement. The system's overall efficiency peaked at 7.32%, which, while lower than conventional wind turbines, represents good performance for a VIV-based system with inherent advantages in terms of simplicity and operational characteristics.

Future work should focus on optimizing the spring-mass-damper parameters to broaden the effective wind speed range and exploring adaptive tuning mechanisms to maintain resonance across varying wind conditions. Additionally, investigations into alternative PMLG designs could further increase electrical power output and system efficiency.

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